Appendix

A. Proof of Lemma 1

Proof. Denote

$$\mathbf{w}_{i_k} = \mathrm{vec}\left(\mathbf{W}_{i_k}\right) \in \mathbb{C}^{N_k^t N_{i_k}^s}, \quad \tilde{\mathbf{A}}_{i_k} = \begin{bmatrix} \mathbf{A}_k & \cdots & \mathbf{0} \\ \vdots & \ddots & \vdots \\ \mathbf{0} & \cdots & \mathbf{A}_k \end{bmatrix} \in \mathbb{C}^{N_k^t N_{i_k}^s \times N_k^t N_{i_k}^s},$$

and we have

$$\begin{aligned} f_1\left(\mathbf{W}\right) &= f\left(\mathbf{W}^t\right) - \sum_{k=1}^{K}\sum_{i=1}^{I_k}\Bigg[\mathrm{tr}\left(\left(\mathbf{W}_{i_k} - \mathbf{W}_{i_k}^t\right)^{\mathsf{H}} \mathbf{A}_k^t \left(\mathbf{W}_{i_k} - \mathbf{W}_{i_k}^t\right)\right) \\ &\qquad + 2\mathrm{Re}\left(\mathrm{tr}\left(\left(\mathbf{G}_{i_k}^t\right)^{\mathsf{H}}\left(\mathbf{W}_{i_k} - \mathbf{W}_{i_k}^t\right)\right)\right)\Bigg] \\ &= f\left(\mathbf{W}^t\right) - \sum_{k=1}^{K}\sum_{i=1}^{I_k}\Bigg[\left(\mathbf{w}_{i_k} - \mathbf{w}_{i_k}^t\right)^{\mathsf{H}} \tilde{\mathbf{A}}_{i_k}\left(\mathbf{w}_{i_k} - \mathbf{w}_{i_k}^t\right) \\ &\qquad + 2\mathrm{Re}\left(\mathrm{vec}\left(\mathbf{G}_{i_k}^t\right)^{\mathsf{H}}\left(\mathbf{w}_{i_k} - \mathbf{w}_{i_k}^t\right)\right)\Bigg] \\ &= f\left(\mathbf{W}^t\right) - \left(\mathrm{vec}\left(\mathbf{W}\right) - \mathrm{vec}\left(\mathbf{W}^t\right)\right)^{\mathsf{H}} \mathbf{A}\left(\mathrm{vec}\left(\mathbf{W}\right) - \mathrm{vec}\left(\mathbf{W}^t\right)\right) \\ &\qquad + 2\mathrm{Re}\left(\mathrm{vec}\left(\mathbf{G}\right)^{\mathsf{H}}\left(\mathrm{vec}\left(\mathbf{W}\right) - \mathrm{vec}\left(\mathbf{W}^t\right)\right)\right), \end{aligned}$$

where

$$\mathbf{A} = \begin{bmatrix} \begin{matrix} \tilde{\mathbf{A}}_{1_1} & \cdots & \mathbf{0} \\ \vdots & \ddots & \vdots \\ \mathbf{0} & \cdots & \tilde{\mathbf{A}}_{I_{1_1}} \end{matrix} & & \\ & \ddots & \\ & & \begin{matrix} \tilde{\mathbf{A}}_{1_K} & \cdots & \mathbf{0} \\ \vdots & \ddots & \vdots \\ \mathbf{0} & \cdots & \tilde{\mathbf{A}}_{I_{K_K}} \end{matrix} \end{bmatrix},$$

and $\mathbf{G} = [\mathbf{G}_{1_1}, \ldots, \mathbf{G}_{I_{1_1}}, \ldots, \mathbf{G}_{1_K}, \ldots, \mathbf{G}_{I_{K_K}}]$. Denote the matrix

$$\mathbf{M}_{i_k} = \left(\mathbf{F}_{i_k}^t + \mathbf{H}_{i_k,k}\mathbf{W}_{i_k}^t\left(\mathbf{W}_{i_k}^t\right)^{\mathsf{H}}\mathbf{H}_{i_k,k}^{\mathsf{H}}\right)^{-1}\mathbf{H}_{i_k,k}\mathbf{W}_{i_k}^t\left(\mathbf{W}_{i_k}^t\right)^{\mathsf{H}}\mathbf{H}_{i_k,k}^{\mathsf{H}}\left(\mathbf{F}_{i_k}^t\right)^{-1},$$

where $i = 1, \ldots, I_k$ and $k = 1, \ldots, K$. Denote the matrix

$$\mathbf{M}_{i_k} = \left(\mathbf{F}_{i_k}^t + \mathbf{H}_{i_k,k}\mathbf{W}_{i_k}^t\left(\mathbf{W}_{i_k}^t\right)^{\mathsf{H}}\mathbf{H}_{i_k,k}^{\mathsf{H}}\right)^{-1}\mathbf{H}_{i_k,k}\mathbf{W}_{i_k}^t\left(\mathbf{W}_{i_k}^t\right)^{\mathsf{H}}\mathbf{H}_{i_k,k}^{\mathsf{H}}\left(\mathbf{F}_{i_k}^t\right)^{-1},$$

where $i = 1, \ldots, I_k$ and $k = 1, \ldots, K$. And $\mathbf{M}_{i_k}$ can be rewritten in a symmetric form

$$\begin{aligned} \mathbf{M}_{i_k} &= \left(\mathbf{F}_{i_k}^t + \mathbf{H}_{i_k,k}\mathbf{W}_{i_k}^t\left(\mathbf{W}_{i_k}^t\right)^{\mathsf{H}}\mathbf{H}_{i_k,k}^{\mathsf{H}}\right)^{-1}\mathbf{H}_{i_k,k}\mathbf{W}_{i_k}^t\left(\mathbf{W}_{i_k}^t\right)^{\mathsf{H}}\mathbf{H}_{i_k,k}^{\mathsf{H}}\left(\mathbf{F}_{i_k}^t\right)^{-1} \\ &= \left(\mathbf{F}_{i_k}^t + \mathbf{H}_{i_k,k}\mathbf{W}_{i_k}^t\left(\mathbf{W}_{i_k}^t\right)^{\mathsf{H}}\mathbf{H}_{i_k,k}^{\mathsf{H}}\right)^{-1}\left(\mathbf{F}_{i_k}^t + \mathbf{H}_{i_k,k}\mathbf{W}_{i_k}^t\left(\mathbf{W}_{i_k}^t\right)^{\mathsf{H}}\mathbf{H}_{i_k,k}^{\mathsf{H}} - \mathbf{F}_{i_k}^t\right)\left(\mathbf{F}_{i_k}^t\right)^{-1} \\ &= \left(\mathbf{F}_{i_k}^t\right)^{-1} - \left(\mathbf{F}_{i_k}^t + \mathbf{H}_{i_k,k}\mathbf{W}_{i_k}^t\left(\mathbf{W}_{i_k}^t\right)^{\mathsf{H}}\mathbf{H}_{i_k,k}^{\mathsf{H}}\right)^{-1}. \end{aligned} \tag{10}$$

Let $\underline{\mathbf{N}}_{j_l} = \sum_{(i,k)\neq(j,l)} \mathbf{H}_{j_l,k}\mathbf{W}_{i_k}^t \left(\mathbf{W}_{i_k}^t\right)^{\mathsf{H}} \mathbf{H}_{j_l,k}^{\mathsf{H}} \succeq \mathbf{0}$, according to the matrix inversion lemma, we have

$$
\begin{aligned}
\mathbf{A}_k &= \sum_{(j,l)} \alpha_{j_l} \mathbf{H}_{j_l,k}^{\mathsf{H}} \mathbf{M}_{j_l} \mathbf{H}_{j_l,k} \\
&= \sum_{(j,l)} \alpha_{j_l} \mathbf{H}_{j_l,k}^{\mathsf{H}} \left(\left(\mathbf{F}_{j_l}^t\right)^{-1} - \left(\mathbf{F}_{j_l}^t + \mathbf{H}_{j_l,l}\mathbf{W}_{j_l}^t \left(\underline{\mathbf{W}}_{j_l}^t\right)^{\mathsf{H}} \mathbf{H}_{j_l,l}^{\mathsf{H}} \right)^{-1} \right) \mathbf{H}_{j_l,k} \\
&\preceq \sum_{(j,l)} \alpha_{j_l} \mathbf{H}_{j_l,k}^{\mathsf{H}} \left(\mathbf{F}_{j_l}^t\right)^{-1} \mathbf{H}_{j_l,k}, \\
&= \sum_{(j,l)} \alpha_{j_l} \mathbf{H}_{j_l,k}^{\mathsf{H}} \left(\sigma_{j_l}^2 \mathbf{I} + \underline{\mathbf{N}}_{j_l}\right)^{-1} \mathbf{H}_{j_l,k} \\
&= \sum_{(j,l)} \alpha_{j_l} \mathbf{H}_{j_l,k}^{\mathsf{H}} \left(\frac{1}{\sigma_{j_l}^2}\mathbf{I} - \frac{1}{\sigma_{j_l}^2} \underline{\mathbf{N}}_{j_l}^{1/2} \left(\mathbf{I} + \frac{1}{\sigma_{j_l}^2}\underline{\mathbf{N}}_{j_l} \right)^{-1} \underline{\mathbf{N}}_{j_l}^{1/2} \frac{1}{\sigma_{j_l}^2} \right) \mathbf{H}_{j_l,k} \\
&\preceq \sum_{(j,l)} \frac{\alpha_{j_l}}{\sigma_{j_l}^2} \mathbf{H}_{j_l,k}^{\mathsf{H}} \mathbf{H}_{j_l,k} \preceq \sum_{(j,l)} \frac{\alpha_{j_l}}{\sigma_{j_l}^2} \lambda_{\max}\left(\mathbf{H}_{j_l,k}^{\mathsf{H}}\mathbf{H}_{j_l,k}\right) \mathbf{I} = \left(\sum_{(j,l)} \frac{\alpha_{j_l}}{\sigma_{j_l}^2} \left\|\mathbf{H}_{j_l,k}\right\|_2^2 \right) \mathbf{I},
\end{aligned}
$$

for $k = 1, \ldots, k$. Denote $L = \max_k \sum_{(j,l)} \frac{2\alpha_{j_l}}{\sigma_{j_l}^2} \left\|\mathbf{H}_{j_l,k}\right\|_2^2$, and the Hessian of the WSR can be bounded as

$$
\nabla^2 f\left(\mathbf{W}^t\right) \succeq -\mathbf{A} \succeq -L\mathbf{I}, \tag{11}
$$

for all $\mathbf{W}_{i_k}^t \in \mathbb{C}^{N_t \times N_s}$, which implies the gradient of f is L-Lipchitz. □

B. Proof of Lemma 3

Proof. Let $\tilde{\mathbf{F}}_{i_k} = \mathbf{F}_{i_k} + \mathbf{H}_{i_k,k}\mathbf{W}_{i_k}\mathbf{W}_{i_k}^{\mathsf{H}}\mathbf{H}_{i_k,k}^{\mathsf{H}} = \sum_{(j,l)} \mathbf{H}_{i_k,l}\mathbf{W}_{j_l}\mathbf{W}_{j_l}^{\mathsf{H}}\mathbf{H}_{i_k,l}^{\mathsf{H}} + \sigma_{i_k}^2\mathbf{I}$, the gradient $\nabla_{\mathbf{W}_{i_k}} f\left(\{\mathbf{W}_{i_k}\}\right) = \mathbf{G}_{i_k}$ can be rewritten as follows:

$$
\begin{aligned}
\mathbf{G}_{i_k} &= \left(\alpha_{i_k}\mathbf{H}_{i_k,k}^{\mathsf{H}}\mathbf{F}_{i_k}^{-1}\mathbf{H}_{i_k,k} - \sum_{(j,l)} \alpha_{j_l}\mathbf{H}_{j_l,k}^{\mathsf{H}} \left(\mathbf{F}_{j_l}^{-1} - \tilde{\mathbf{F}}_{j_l}^{-1}\right) \mathbf{H}_{j_l,k} \right) \mathbf{W}_{i_k} \\
&= \left(\sum_{(j,l)} \alpha_{j_l}\mathbf{H}_{j_l,k}^{\mathsf{H}}\tilde{\mathbf{F}}_{j_l}^{-1}\mathbf{H}_{j_l,k} - \sum_{(j,l)\neq(i,k)} \alpha_{j_l}\mathbf{H}_{j_l,k}^{\mathsf{H}}\mathbf{F}_{j_l}^{-1}\mathbf{H}_{j_l,k} \right) \mathbf{W}_{i_k}.
\end{aligned} \tag{12}
$$

Considering the sum of the inner products between $\mathbf{G}_{i_k}$ and $\mathbf{W}_{i_k}$, i.e.

$$
\begin{aligned}
\sum_{(i,k)} \operatorname{tr}\left(\mathbf{W}_{i_k}^{\mathsf{H}}\mathbf{G}_{i_k}\right) &= \sum_{(i,k)} \operatorname{tr}\left(\mathbf{W}_{i_k}^{\mathsf{H}} \left(\sum_{(j,l)} \alpha_{j_l}\mathbf{H}_{j_l,k}^{\mathsf{H}}\tilde{\mathbf{F}}_{j_l}^{-1}\mathbf{H}_{j_l,k} - \sum_{(j,l)\neq(i,k)} \alpha_{j_l}\mathbf{H}_{j_l,k}^{\mathsf{H}}\mathbf{F}_{j_l}^{-1}\mathbf{H}_{j_l,k} \right) \mathbf{W}_{i_k} \right) \\
&= \sum_{(j,l)} \alpha_{j_l} \operatorname{tr}\left(\sum_{(i,k)} \mathbf{W}_{i_k}^{\mathsf{H}}\mathbf{H}_{j_l,k}^{\mathsf{H}}\tilde{\mathbf{F}}_{j_l}^{-1}\mathbf{H}_{j_l,k}\mathbf{W}_{i_k} - \sum_{(i,k)\neq(j,l)} \mathbf{W}_{i_k}^{\mathsf{H}}\mathbf{H}_{j_l,k}^{\mathsf{H}}\mathbf{F}_{j_l}^{-1}\mathbf{H}_{j_l,k}\mathbf{W}_{i_k} \right) \\
&= \sum_{(j,l)} \alpha_{j_l} \operatorname{tr}\left(\tilde{\mathbf{F}}_{j_l}^{-1} \left(\sum_{(i,k)} \mathbf{H}_{j_l,k}\mathbf{W}_{i_k}\mathbf{W}_{i_k}^{\mathsf{H}}\mathbf{H}_{j_l,k}^{\mathsf{H}} \right) - \mathbf{F}_{j_l}^{-1} \left(\sum_{(i,k)\neq(j,l)} \mathbf{H}_{j_l,k}\mathbf{W}_{i_k}\mathbf{W}_{i_k}^{\mathsf{H}}\mathbf{H}_{j_l,k}^{\mathsf{H}} \right) \right).
\end{aligned} \tag{13}
$$

Since matrices

$$
\begin{aligned}
\tilde{\mathbf{F}}_{i_k} &= \sum_{(j,l)} \mathbf{H}_{i_k,l}\mathbf{W}_{j_l}\mathbf{W}_{j_l}^{\mathsf{H}}\mathbf{H}_{i_k,l}^{\mathsf{H}} + \sigma_{i_k}^2\mathbf{I}, \\
\mathbf{F}_{i_k} &= \sum_{(j,l)\neq(i,k)} \mathbf{H}_{i_k,l}\mathbf{W}_{j_l}\mathbf{W}_{j_l}^{\mathsf{H}}\mathbf{H}_{i_k,l}^{\mathsf{H}} + \sigma_{i_k}^2\mathbf{I},
\end{aligned} \tag{14}
$$

for $i=1,\ldots,I_k$ and $k=1,\ldots,K$, according to matrix inversion lemma, we have

$$\begin{aligned}
\sum_{(i,k)} \operatorname{tr}\left(\mathbf{W}_{i_k}^{\mathsf{H}} \mathbf{G}_{i_k}\right) &= \sum_{(j,l)} \alpha_{j_l} \sigma_{j_l}^2 \operatorname{tr}\left(\mathbf{F}_{j_l}^{-1} - \tilde{\mathbf{F}}_{j_l}^{-1}\right) \\
&= \sum_{(j,l)} \alpha_{j_l} \sigma_{j_l}^2 \operatorname{tr}\left(\mathbf{F}_{j_l}^{-1} - \left(\mathbf{F}_{j_l} + \mathbf{H}_{j_l,l} \mathbf{W}_{j_l} \mathbf{W}_{j_l}^{\mathsf{H}} \mathbf{H}_{j_l,l}^{\mathsf{H}}\right)^{-1}\right) \\
&= \sum_{(j,l)} \alpha_{j_l} \sigma_{j_l}^2 \operatorname{tr}\left(\mathbf{F}_{j_l}^{-1} - \left(\mathbf{F}_{j_l}^{-1} - \mathbf{F}_{j_l}^{-1} \mathbf{H}_{j_l,l} \mathbf{W}_{j_l} \left(\mathbf{I} + \mathbf{W}_{j_l}^{\mathsf{H}} \mathbf{H}_{j_l,l}^{\mathsf{H}} \mathbf{F}_{j_l}^{-1} \mathbf{H}_{j_l,l} \mathbf{W}_{j_l}\right)^{-1} \mathbf{W}_{j_l}^{\mathsf{H}} \mathbf{H}_{j_l,l}^{\mathsf{H}} \mathbf{F}_{j_l}^{-1}\right)\right) \\
&= \sum_{(j,l)} \alpha_{j_l} \sigma_{j_l}^2 \operatorname{tr}\left(\mathbf{F}_{j_l}^{-1} \mathbf{H}_{j_l,l} \mathbf{W}_{j_l} \left(\mathbf{I} + \mathbf{W}_{j_l}^{\mathsf{H}} \mathbf{H}_{j_l,l}^{\mathsf{H}} \mathbf{F}_{j_l}^{-1} \mathbf{H}_{j_l,l} \mathbf{W}_{j_l}\right)^{-1} \mathbf{W}_{j_l}^{\mathsf{H}} \mathbf{H}_{j_l,l}^{\mathsf{H}} \mathbf{F}_{j_l}^{-1}\right) \\
&\geq 0,
\end{aligned} \tag{15}$$

the inequality holds if and only if $\mathbf{H}_{i_k,k} \mathbf{W}_{i_k} = \mathbf{0}$ for all $i=1,\ldots,I_k$, $k=1,\ldots,K$. □

C. Proof of Theorem 4

Proof. If the gradient zero, the inner product

$$\sum_{(i,k)} \operatorname{tr}\left(\mathbf{W}_{i_k}^{\mathsf{H}} \mathbf{G}_{i_k}\right) = 0, \quad \forall \mathbf{W}_{i_k} \in \mathbb{C}^{N_k^t \times N_{i_k}^s}.$$

According to (15), for all $j=1,\ldots,I_l$, $l=1,\ldots,K$ we have

$$\begin{aligned}
& \operatorname{tr}\left(\mathbf{F}_{j_l}^{-1} \mathbf{H}_{j_l,l} \mathbf{W}_{j_l} \left(\mathbf{I} + \mathbf{W}_{j_l}^{\mathsf{H}} \mathbf{H}_{j_l,l}^{\mathsf{H}} \mathbf{F}_{j_l}^{-1} \mathbf{H}_{j_l,l} \mathbf{W}_{j_l}\right)^{-1} \mathbf{W}_{j_l}^{\mathsf{H}} \mathbf{H}_{j_l,l}^{\mathsf{H}} \mathbf{F}_{j_l}^{-1}\right) = 0 \\
\iff & \left\| \mathbf{F}_{j_l}^{-1} \mathbf{H}_{j_l,l} \mathbf{W}_{j_l} \left(\mathbf{I} + \mathbf{W}_{j_l}^{\mathsf{H}} \mathbf{H}_{j_l,l}^{\mathsf{H}} \mathbf{F}_{j_l}^{-1} \mathbf{H}_{j_l,l} \mathbf{W}_{j_l}\right)^{-1/2} \right\|_F = 0 \\
\iff & \mathbf{F}_{j_l}^{-1} \mathbf{H}_{j_l,l} \mathbf{W}_{j_l} \left(\mathbf{I} + \mathbf{W}_{j_l}^{\mathsf{H}} \mathbf{H}_{j_l,l}^{\mathsf{H}} \mathbf{F}_{j_l}^{-1} \mathbf{H}_{j_l,l} \mathbf{W}_{j_l}\right)^{-1/2} = \mathbf{0} \\
\iff & \mathbf{H}_{j_l,l} \mathbf{W}_{j_l} = \mathbf{0},
\end{aligned}$$

and vice versa. □

D. Proof of Corollary 5

Proof. Assume the optimal beamformers of the problem (2) satisfy $\left\{\mathbf{W}_{i_k}^{\star}\right\} \in \mathring{\mathcal{C}}$. According to the first order condition, we have $\mathbf{G}_{i_k} = \mathbf{0}$ for all $i=1,\ldots,I_k$, $k=1,\ldots,K$. According to Theorem 4, we have $\mathbf{H}_{i_k,k} \mathbf{W}_{i_k}^{\star} = \mathbf{0}$. Consider the WSR of the optimal beamformers

$$f\left(\mathbf{W}^{\star}\right) = \sum_{k=1}^{K} \sum_{i=1}^{I_k} \alpha_{i_k} \log\det\left(\mathbf{I} + \left(\mathbf{W}_{i_k}^{\star}\right)^{\mathsf{H}} \mathbf{H}_{i_k,k}^{\mathsf{H}} \mathbf{F}_{i_k}^{-1} \mathbf{H}_{i_k,k} \mathbf{W}_{i_k}^{\star}\right) = 0,$$

which implies the beamformers are not the optimal solution of the problem (2), contradiction. □